

Abdullah Shadek Fahim

+8801816-357640 ✉ asfahimbd@gmail.com in abdullah-fahim7640

Cox's Bazar, Chittagong-4700, Bangladesh

RESEARCH INTERESTS

Machine Learning for Semiconductor Process Modeling, Ion Implantation, VLSI and Compound Semiconductor Devices (SiC, GaAs), Molecular Dynamics, DFT, AI-driven Biomedical Instrumentation, Embedded Systems for Healthcare Applications

EDUCATION

- **Jashore University of Science and Technology (JUST)** 2020 – 2026
Bachelor of Science in Electrical and Electronic Engineering Jashore, Bangladesh
 - **CGPA: 3.80 / 4.00**
 - **Thesis:** *A Dual-Stage Hybrid Random Forest Framework for Forward and Inverse Modeling of Ion Implantation in Si, SiC, and GaAs Substrates.*
 - **Relevant Coursework:** Semiconductor Devices, Power System, VLSI Design, Digital Signal Processing, Microprocessors & Embedded Systems, Power Electronics, Control Systems, Analog Electronics, Engineering Mathematics.
- **Ispahani Public School & College** 2017 – 2019
Higher Secondary Certificate (Science) Chittagong, Bangladesh
 - **GPA: 5.00 / 5.00**

RESEARCH EXPERIENCE

- **Research Assistant** 2025 – Present
Interdisciplinary Laboratory of Advanced Material Processing and Simulation (ILAMPS), JUST
- **Undergraduate Thesis Research** 2025 – 2026
Dept. of EEE, Jashore University of Science and Technology (JUST)
 - Developed a **dual-stage hybrid machine learning framework** using Random Forest to model ion implantation physics across three semiconductor substrates: Silicon (Si), Silicon Carbide (SiC), and Gallium Arsenide (GaAs).
 - Designed a **gatekeeper-constrained architecture** that enforces physical consistency between the forward model (predicting ion range and damage profiles) and the inverse model (inferring implantation parameters from desired profiles).

PUBLICATIONS

C=CONFERENCE, J=JOURNAL

- [J.1] **Abdullah Shadek Fahim**, Musavvir Haque, Abu Taher, Jahedul Islam, Md. Mehedi Hasan, and Md. Amzad Hossain (2026). **Emulating Binary Collision Approximation Physics via a Gatekeeper Constrained Dual-Stage Random Forest: Multi-Output Ion Range and Damage Prediction Across Si, 4H-SiC, and GaAs Substrates.**
Under Review since May 2026: *Computational Materials Science*
- [J.2] Musavvir Haque, **Abdullah Shadek Fahim**, Saklain Mustak Nure Modina Sathi, Jahedul Islam, Md. Tariquzzaman and Md. Amzad Hossain (2026). **Hydrogen Implantation and Diffusion in BCC Tungsten: A Systematic Molecular Dynamics Study over 5–300eV and 300-1500K.**
Under Review since May 2026: *Fusion Engineering and Design*
- [C.1] **Abdullah Shadek Fahim**, Jahedul Islam, Irin Akter Mim, and Nure Modina Sathi (2025). **CardioPredictor: A Real-time Non-Invasive and Cuffless Cardiovascular Disease Prediction System using 10-Fold Cross Validation on the Framingham Dataset.**
DOI: [10.1109/QPAIN66474.2025.11171821](https://doi.org/10.1109/QPAIN66474.2025.11171821) Publisher: *IEEE Xplore*, September 2025.
- [C.2] **Abdullah Shadek Fahim**, Abu Taher, Musavvir Haque, and Md. Amzad Hossain (2026). **A Validated Machine Learning Surrogate for Rapid Forward and Reverse Ion Implantation Modeling in Gallium Arsenide Substrates for Plasma Based Semiconductor Industry.**
Accepted at: 2026 IEEE International Conference on Plasma Science (ICOPS), USA.

PROJECTS

Research & Design Projects

- **Varley Loop Test Kit:** Designed and implemented a functional test kit for locating underground cable faults, serving as a practical tool for the university's analog electronics lab.
- **Line Fault Detection Circuit:** Designed a line-to-line fault detection and temperature protection system for switchgear applications.

Embedded Systems & Robotics

- **Self-Balancing Robot:** Developed a two-wheel self-balancing robot using Arduino Nano and MPU6050, implementing a PID control algorithm for stable, real-time balancing.
- **Line Following Robot:** Developed an autonomous robot using Arduino Nano and PID control, implementing a 5-channel IR sensor array for precise, real-time path tracking.
- **Design and Development of Astable Multivibrator:** Designed the circuit on NI Multisim, then fabricated the hardware using a 555 timer IC.

PROFESSIONAL TRAINING & FIELD VISITS

- **Department of EEE, Daffodil International University (DIU)** *April 2026*
Training – Solar System Planning, Design, Installation, and Operation Savar, Bangladesh
 - Completed a 32-hour hands-on training on PVSyst for solar design and live installation of a 5 kW off-grid hybrid solar system at the DIU Solar System Lab.
- **Training Institute for Chemical Industries (TICI)** *Jun 2025 – Jul 2025*
Industrial Training – Instrumentation and Control Systems Ghorashal, Bangladesh
 - Completed intensive training in Instrumentation, PLC, DCS, and Electrical Systems. Achieved **A+ grade**.
- **Ulkasemi Pvt. Ltd.** *May 2025*
Training – VLSI Design (Analog IC & Physical Design) Dhaka, Bangladesh
 - Completed training on analog IC design, custom layout methodology, and place-and-route in an industry setting.
- **Khulna 330 MW Dual Fuel Combined Cycle Power Plant** *October 2025*
Bangladesh Power Development Board Khulna, Bangladesh
 - An intensive, full-day site visit to an operational gas and diesel-based power plant—gaining hands-on insights from the central control room to the boilers, turbines, and auxiliary systems.

LEADERSHIP EXPERIENCE

- **Mentor, IEEE JUST Student Branch** *December 2025 – Present*
Jashore University of Science and Technology (JUST)
 - Providing strategic guidelines and mentorship to the newly elected executive committee.
 - Advising on operational planning and ensuring sustainability of branch activities.
- **Chair, IEEE JUST Student Branch** *September 2024 – December 2025*
Jashore University of Science and Technology (JUST)
 - Led all strategic, technical, and community initiatives of the branch, overseeing a team of 20+ active members.
 - Organized events, drove membership growth, and coordinated collaborations with IEEE Bangladesh Section.
 - Served as Organizing Chair of the IEEE JUST Signal Processing Student Branch Chapter and the IEEE JUST Women in Engineering Affinity Group.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, MATLAB, Verilog, HDL
- **Machine Learning:** Scikit-learn, NumPy, Pandas, Matplotlib; experience with Random Forest, ensemble methods, cross-validation
- **EDA & Simulation:** Proteus, NI Multisim, ModelSim, PowerFactory, SRIM, Materials Studio, AutoCAD
- **Hardware & Embedded:** Arduino, ESP8266, ESP32, MPU6050, PLC, DCS
- **Documentation & Tools:** LaTeX, Microsoft Office, Adobe InDesign
- **Soft Skills:** Research Communication, Technical Writing, Leadership, Mentorship, Public Speaking

HONORS AND AWARDS

- **Outstanding Student Volunteer Award 2025 (Honorable Mention)** December 2025
IEEE Bangladesh Section
 - Recognized for exceptional leadership and contributions to IEEE activities throughout 2025.
- **2nd Runner-Up – IEEE Bangladesh Section SYW Idea Pitching Competition** 2025
IEEE Bangladesh Section
 - Recognized for the Presentation of an AI-Based Skin Transplantation System for Burn Patients: A National Sync Database.
- **Keynote Speaker – From Idea to Publication: Writing a Conference Paper** 2025
IEEE JUST Student Branch Webinar
 - Conducted a comprehensive session on the full pipeline of writing and submitting an IEEE conference paper, from problem formulation to publication.

VOLUNTEER EXPERIENCE

- **Lead Volunteer – ETSD 2026** May 2026
Faculty of Engineering and Technology, JUST
 - Served as lead volunteer at the 2nd International Conference on Engineering and Technology for Sustainable Development, Jashore, Bangladesh.
 - Co-Editor of the Conference Proceeding
- **Lead Coordinator - STUDENT-LED CON 2025** December 2025
IEEE Bangladesh Section
 - Co-organized the International Student-Led Conference 2025, held in hybrid mode.
- **Volunteer – IEEE ICCIT 2024** December 2024
IEEE Bangladesh Section
 - Provided organizational support for the 27th IEEE International Conference on Computer and Information Technology, Cox's Bazar, Bangladesh.

PROFESSIONAL MEMBERSHIPS

- **IEEE (Institute of Electrical and Electronics Engineers)** 2023 – Present
Student Member – Membership ID: 100102890

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (Professional Working Proficiency)

Other: Freelance book formatting, graphic and Website design professional (Fiverr Level 2 Seller; 350 completed projects, Overall 4.9-star rating since 2021).

REFERENCES

Md. Amzad Hossain, Dr. Eng.
Associate Professor, Dept. of EEE
Jashore University of Science and
Technology (JUST)
Counselor, IEEE JUST Student
Branch
Email: mahossain.just@gmail.com
Relationship: Academic Supervisor

Dr. Mehedi Hassan (Jewel)
Associate Professor, Dept. of EEE
Jashore University of Science and
Technology (JUST)
Email: m.hassan@just.edu.bd
Relationship: Academic Advisor

Professor Dr. Engr. Imran Khan
Chairman, Dept. of EEE &
Dean, Faculty of Engineering and
Technology
Jashore University of Science and
Technology (JUST)
Email: I.khan@just.edu.bd
Relationship: Head of Department